

Marie-Christine Perry

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Product leader with hands-on LLM and Generative AI experience and a life-sciences research background in immunology and biologics. Proven track record of shipping production AI systems end-to-end for enterprise pharma customers including Amgen, Merck, Sanofi, GSK, and Pfizer, from problem discovery through ML evaluation, launch, and iteration. Combines strong technical depth in Python, RAG architectures, prompt engineering, and LLM evals with deep domain fluency in scientific content workflows and regulated life-sciences data. Actively deepening production AI skills through a Deploying AI credential to keep pace with the fast-moving LLM stack.

CORE COMPETENCIES

AI/ML & Generative AI: LLM-based classification and tagging · Prompt engineering · RAG architectures · ML evaluation frameworks (accuracy, precision/recall, latency, throughput) · Task decomposition · AI agent design · Vertex AI · Production ML pipelines

Product Leadership: End-to-end product ownership · 0 to 1 product development · Roadmap strategy and prioritization · Enterprise stakeholder management · Success metrics and KPIs · User research and discovery · Spec writing · Agile/Scrum · Certified Product Owner

Technical: Python · SQL · R · BigQuery · GCP · ETL pipelines · Looker dashboards · Prefect workflows · Internal APIs · Git

Life Sciences Domain: Scientific literature and publication data · Immunology · Biologics workflows · Multimodal R&D data · Regulated content review · ELN/LIMS and lab instrument integration · Peer-reviewed publishing (3 co-authored papers)

PROFESSIONAL EXPERIENCE

Product Manager, Luma Adaptive Workflows | Dotmatics, Toronto Feb 2026 to Present

Luma is Dotmatics' AI-powered scientific R&D platform for pharma and life-sciences customers.

- **Own product direction for Adaptive Workflows**, enabling customers to build digital twins of physical lab workflows within the Luma platform.
- **Partner with the Luma AI agent team** to integrate adaptive workflow functionality into agentic capabilities, translating complex scientific workflows into product requirements and high-impact AI use cases.

Product Owner & Science Tech Lead | BenchSci, Toronto 2022 to 2026

AI-powered SaaS platform serving scientists at the world's largest pharma companies. Owned AI product areas end-to-end, leading cross-functional teams of engineers, ML engineers, data scientists, and designers.

- **Conceived and owned Corpus Sorter, an LLM-based relevance classification system** that curated ~12M high-signal scientific publications, replacing a noisy rules-based ingestion pipeline. Defined the problem, validated the approach with pharma users, wrote specs, partnered with engineers to productionize prompts, and shipped iteratively to establish the scalable foundation for multiple downstream products.
- **Designed and ran ML evaluation frameworks** across accuracy, precision/recall, latency, and throughput to ensure Corpus Sorter was production-ready and defensible, enabling senior-leadership confidence and scalable deployment across 11 therapeutic areas.
- **Drove senior-leadership buy-in for a relevance-first corpus strategy** by articulating tradeoffs, presenting performance data, and defining a clear migration path, shifting the company's long-term approach to corpus curation and expansion.
- **Authored the long-term corpus expansion roadmap** to close coverage gaps, identifying ~1.5M missing publications and implementing automated acquisition pipelines via Ficstar and PubMed-based requests, unlocking systematic, scalable corpus growth.

- **Established the Therapeutic Area Health framework** across 11 user-aligned therapeutic areas, evolving from MeSH and biomedical ontology classification to LLM-driven approaches, and introduced release-style reporting that became a core input to corpus and product strategy.
- **Built company-wide product observability from scratch** by self-teaching Looker and integrating Prefect workflows, delivering near-real-time dashboards for corpus coverage, relevance, and growth that became the single source of truth for scientists, product, engineering, and leadership.
- **Automated analytics and reporting pipelines** using Python, BigQuery SQL, and internal APIs, reducing manual analysis, scaling analytical capacity across corpus and clinical trial teams without additional headcount, and accelerating iteration cycles.
- **Served as primary scientific and product liaison for enterprise pharma customers** (Amgen, Merck, Sanofi, GSK, Pfizer), delivering data-backed coverage analyses and deep-dive user research that strengthened customer trust, supported renewals, and enabled successful pilots and expansions.
- **Led technical mentorship and enablement**, scoping and guiding a TKS intern project on LLMs for accelerated report writing (Python, LLMs, Colab, Vertex AI), and onboarding team members on corpus and TA tooling to reduce single-point dependency and strengthen product culture.

SCIENTIFIC RESEARCH EXPERIENCE

Researcher, Mucosal Immunology | The Hospital for Sick Children, Toronto 2020 to 2022

- Analyzed high-dimensional immunology datasets (flow cytometry, CyTOF), translating noisy biological signals into reproducible findings and building intuition for the data-driven workflows pharma R&D and medical affairs teams rely on.
- Designed and executed in vivo studies to isolate host and microbiome effects, applying first-principles reasoning to complex biological systems.

MSc Researcher, Applied Immunology | University Health Network, Toronto 2018 to 2020

- Generated scalable datasets using NGS, qPCR, and high-throughput ELISA; co-authored 3 peer-reviewed publications.
- Conducted systematic literature reviews and applied quantitative analysis to interpret experimental outcomes, giving firsthand experience with the MLR-adjacent pain points of regulated scientific content: manual review, reference matching, and compliance against source material.

Research Associate, Protein Engineering | Innovative Targeting Solutions, Vancouver 2017 to 2018

- Supported biologics R&D across antibody engineering, cell culture, purification, and QC workflows directly relevant to pharma commercial and medical affairs content.

EDUCATION & CERTIFICATIONS

Deploying AI Credential, University of Toronto (in progress)	2026
Data Science & Machine Learning Certificate, University of Toronto	2026
Certified Product Owner, Scrum Alliance	2025
MSc, Applied Immunology, University of Toronto	2020
BSc, Biochemistry, University of Victoria	2017

ADDITIONAL LEADERSHIP

- Co-Chair, QueerSci Employee Resource Group and BenchShe+ ambassador (BenchSci): Led cross-functional initiatives supporting inclusive community-building and knowledge sharing.